

MD Azizul Hakim

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🐙 [Github](#)

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🎓 [Google Scholar](#)

🌐 [Personal Website](#)

RESEARCH INTERESTS

Large Language Models, Reinforcement Learning, Test-Time Compute Efficiency in Large Language Models, Training Quality and Deployment Behaviour of LLMs, Inference Optimization of LLMs

EDUCATION

Diploma in Computer Science and Technology

Jul 2021 – Feb 2026

Bangladesh Sweden Polytechnic Institute

CGPA: 3.40/4.0

Seeking fully funded BSc opportunities at international universities for 2026–2027 admission, with research profile as primary differentiator.

EXPERIENCE

AI Researcher Intern at Universe Soft Tech

Dec 2025 – Feb 2026

PUBLICATIONS

Top-Tier Venue — Sole Author [1] Hakim, MD Azizul (2026). “Training Quality Determines Efficiency Boundaries in Test-Time Reasoning.” *Neural Information Processing Systems (NeurIPS 2026 Accepted Poster - Main Track)*. 50 models, 67,000+ evaluations, 6 benchmarks. [\[code\]](#)

Journal Articles

Journal Articles [J1] Hakim, M.A., & Alam, M.I. (2025). “Biologically inspired neural network layer with homeostatic regulation and adaptive repair mechanisms.” *Scientific Reports* (Nature Portfolio), 15, 9114. [\[paper\]](#) [\[code\]](#)

Conference Papers [C1] Hakim, M.A., Ifty, R.A., Delowar, K.E., Shuvo, S.A., Tasneem, Z., & Alam, M.I. (2025). “A Novel Neuroplasticity-Based Approach for Enhanced Generalization in Computer Vision.” *IEEE Conference Proceedings*. [\[paper\]](#)

[C2] **Hakim, M.A.,** Ifty, R.A., Delowar, K.E., Chowdhury, S.H., Rashid, I., & Shakib, M. (2025). “Nutri-Guard: LLM-Driven Nutritional Assessment for Chronic Disease Prevention.” *IEEE Conference Proceedings*. [\[paper\]](#)

[C3] **Hakim, M.A.** (2025). “Neuroadaptive Intelligence: A Biologically-Inspired, Self-Repairing Neural Layer with Adaptive Learning for Molecular Graph Classification.” *IEEE Conference Proceedings*. [\[paper\]](#)

Preprints [P1] Hakim, MD Azizul (2025). “Brevity Constraints Reverse Performance Hierarchies in Language Models.” *arXiv preprint*. [\[arXiv\]](#) [\[code\]](#)

[P2] **Hakim, MD Azizul** (2025). “Multi-Scale Temporal Homeostasis Enables Efficient and Robust Neural Networks.” *arXiv preprint*. [\[arXiv\]](#) [\[code\]](#)

[P3] **Hakim, MD Azizul** (2025). “Cross-Domain Generalization Failure in Lightweight Intrusion Detection Models for IIoT Networks” *arXiv preprint*. [\[arXiv\]](#) [\[code\]](#)

AWARDS & HONORS

2nd Runner-Up – AI Agent Development Competition

2025

Insten Company in collaboration with International Islamic University Chittagong (IIUC)

Kaggle Notebooks Master — kaggle.com/hakim11

2024

SELECTED PROJECTS

FounderAI – Executive AI Agent Platform

GitHub

- Comprehensive multi-agent system providing startup founders with AI-powered C-suite expertise (CMO, CPO, CTO, COO)
- Implemented multi-agent orchestration using CrewAI and LangGraph with Model Context Protocol (MCP)
- Capabilities spanning market intelligence, technical architecture, and fundraising strategy

LangChain • LangGraph • CrewAI • Multi-Agent Systems

AI-Driven Tourist Spot Finder

GitHub

- Intelligent travel companion with multi-agent architecture for personalized destination recommendations
- FastAPI production backend with ML recommendation engine (collaborative + content-based filtering)
- Integrated LLMs for dynamic destination descriptions and real-time data (weather, pricing, availability)

FastAPI • Machine Learning • LangChain • MongoDB

Health-Conscious Food Recommendation System

GitHub

- Personalized nutrition AI with advanced OCR (Mistral OCR) for food label analysis
- LLM-powered nutritional assessment with RAG for real-time medical data integration
- Recommendation engine tailored to specific health conditions

Computer Vision • OCR • RAG • Healthcare AI

Bengali Text-to-Speech Synthesis

GitHub

- Fine-tuned Pegasus transformer for high-quality Bengali speech synthesis
- Full MLOps pipeline: automated training, deployment, and real-time inference API

Transformers • NLP • MLOps • Bengali Language

TECHNICAL SKILLS

Languages	Python (Expert), MATLAB, SQL, JavaScript
ML/DL Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, OpenCV, NLTK, SpaCy, XGBoost, LightGBM, CatBoost
LLM & Agents	LangChain, LangGraph, CrewAI, RAG, MCP, Fine-tuning (Llama, Pegasus, Mistral)
MLOps & DevOps	Docker, Kubernetes, GitHub Actions, Jenkins, CircleCI, DVC, MLflow
Cloud Platforms	AWS, Azure, GCP
Databases	MongoDB, PostgreSQL, MySQL, Redis, Cassandra
Other	RESTful APIs, Microservices, Agile/Scrum, Feature Engineering, Model Interpretability, Responsible AI